Directional Collapse: One-Way Fine-Tuning Erases the Way Back

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Abstract

Many NLP tasks come in pairs whose training instances are the same data read in two directions: obfuscation and deobfuscation, translation each way, text-to-SQL and SQL-to-text, data-to-text and text-to-data. Fine-tuning on one direction destroys the ability the model already had in the other. We characterise this across ⟨⟨n-domains⟩⟩ paired domains and ⟨⟨n-families⟩⟩ model families, and show three things about it. The loss is disproportionate: the inverse direction falls to ⟨⟨abstract-reverse-after⟩⟩ while held-out general ability moves ⟨⟨abstract-general-cost⟩⟩. It is cheap to prevent: replacing ⟨⟨abstract-knee⟩⟩ of the forward pairs with their own reversal restores ⟨⟨abstract-restored⟩⟩ of the lost capability, at matched instances, sequence tokens and optimizer steps. And what is lost is suppressed rather than erased — recoverable by prompting, by scaling the adapter, by a few reversed examples, and by a closed-form filter on the weight update that uses no data at all. Invertibility bounds the repair, and separates two things usually run together: an inverse that is undetermined cannot be restored at any dose, while one that is merely hard to find can be. A consequence follows for attribution: any method credited with restoring an inverse capability on paired data must be compared against a baseline supplying the same directional content and none of the method.

1 Introduction

Paired data carries a direction. Obfuscating a program rewrites it so that it behaves identically but is hard to read; a model can be trained to perform that rewrite, or to undo it. The two tasks use the same pairs of programs, and training the second only requires swapping which half of each pair is the input. Compilation and decompilation, minification and beautification, translation between two languages, and the two directions of semantic parsing all work this way.

This structure is what makes the failure we study both possible and avoidable. Fine-tuning a model on the forward direction of such a pair degrades its ability in the reverse direction — an ability the base model already had. Nikiema et al. (2025) name this *cognitive specialization* and document it for code obfuscation; Zhu et al. (2024) report the same shape in machine translation, where fine-tuning with English on the target side damages translation into other languages. We take the phenomenon as established in those settings and ask what is true of it in general.

The loss is disproportionate, which is why it is easy to miss. ⟨⟨intro-disproportion-sentence⟩⟩

It is cheap to prevent. ⟨⟨intro-dose-sentence⟩⟩

Invertibility bounds the repair, and it is two properties, not one. An inverse can be *undetermined* — the information needed is not present in the input, and no quantity of reverse data can supply it — or *determined but hard to find*, where the information is there and search is the obstacle. These predict different things about a cure, and §6 separates them with a synthetic ladder that varies determinability directly and a domain that holds determinability fixed while varying search difficulty.

What is lost is suppressed, not erased. ⟨⟨intro-mechanism-sentence⟩⟩

A general test follows, and it needs no new data. For any method that adds instances to paired training data, train the arm that supplies the same directional content and none of the method. This is more than advice because the ablation arm is also the remedy: running it pays for itself whichever way the comparison falls. §8 applies it to ⟨⟨n-attrib-methods⟩⟩ published objectives.

2 Directional collapse, defined

A *paired task* is a pair of datasets (X, Y) and (Y, X) over the same instances, where the reverse training set is obtained from the forward one by exchanging input and target. We call $X \to Y$ the *forward* direction and $Y \to X$ the *reverse*. *Directional collapse* is a fall in reverse-direction success after fine-tuning on forward-direction data alone, measured against the untouched model.

Every domain has a strict per-instance criterion with an automatic check (§3). Two failure modes are tracked separately and never counted as success:

Echo the output reproduces its own input. This is not hygiene. In JSON-to-YAML conversion, JSON is valid YAML, so a model that copies its input *parses* and *compares structurally equal* to the target; without the echo guard the task is not a task. Echo rates rose under forward-only training in earlier work, so a criterion that admits them measures the criterion rather than the capability.

Off-target the output is in the wrong language, format, or notation.

Thresholds for continuous criteria are fixed from the *untouched* model's own score distribution before any fine-tuned model is scored, and are recorded with the date and quantile that produced them.

3 Setup

Domains. ⟨⟨setup-domains-sentence⟩⟩ [table domains not generated yet]

Models. ⟨⟨setup-models-sentence⟩⟩

Arms and budget. The forward-only baseline (SFT) trains on forward pairs. Each MIXk arm replaces $k\%$ of those pairs with their own reversal, partitioned by pair identity so that no pair is seen in both directions. Because the reversal *replaces* rather than adds, instance count, sequence tokens and optimizer steps are matched to SFT at every dose, and the only quantity varying along the ladder is the share of pairs seen backwards. Golovneva et al. (2024) distinguish data-matched from compute-matched reverse training; replacement is matched on both simultaneously, which is what licenses the word "free" in §5 — scoped, there and throughout, to training budget.

Two controls separate directional loss from its alternatives. REPLAY replaces the same share with generic instruction data, so REPLAY − SFT is ordinary forgetting and MIX − REPLAY is what direction specifically buys. REV trains on reverse pairs only and is the ceiling every reverse number is read against; where REV is near zero the direction is not learnable in that cell and no null there is interpretable. [table arms not generated yet]

4 Collapse is general and disproportionate

⟨⟨general-headline-sentence⟩⟩ [table main not generated yet]

It is not ordinary forgetting. ⟨⟨general-replay-contrast⟩⟩

It is not a loss of instruction following. ⟨⟨general-ifeval-contrast⟩⟩

Moderators. ⟨⟨general-moderators⟩⟩

5 A small reversed dose restores it at no cost

⟨⟨dose-headline-sentence⟩⟩ [table dose not generated yet]

Where the knee is. We define the knee as the smallest dose whose reverse success reaches 90% of MIX50's in that cell. ⟨⟨dose-knee-sentence⟩⟩

What it costs. ⟨⟨dose-cost-sentence⟩⟩ Fine-tuning costs general ability at any direction mix; the dose is free relative to the forward-only baseline, not relative to the untouched model.

6 Invertibility bounds what can be restored

⟨⟨invertibility-headline-sentence⟩⟩ [table ladder not generated yet]

Two kinds of hardness. ⟨⟨invertibility-two-kinds⟩⟩

7 Suppressed, not erased

Four instruments, reported as separate verdicts rather than averaged: they measure different things, and a disagreement between them is a finding rather than noise.

Prompting recovers part of what tuning removed. ⟨⟨mech-elicitation⟩⟩

The collapse is a cheap direction in weight space. ⟨⟨mech-alpha⟩⟩

Relearning outruns a capability never held. ⟨⟨mech-relearn⟩⟩ This is also where the contrast with the reversal curse (Berglund et al., 2024) becomes experimental rather than rhetorical: that literature describes capabilities never acquired, and our never-had control is that case, constructed.

A closed-form filter on the update restores it with no data. ⟨⟨mech-spectral⟩⟩

8 Consequences for attribution

⟨⟨attrib-headline-sentence⟩⟩

We do not claim that every auxiliary-objective result on paired data is a directional artifact. We measured ⟨⟨n-attrib-methods⟩⟩; the confound is real, cheap to test for, and currently untested, and anything stronger than that is unsupported.

9 Related work

Four lines of work meet here. Each has a sentence saying what separates it from directional collapse, because in three of the four the distinction is the contribution.

The phenomenon, in single domains. Nikiema et al. (2025) name *cognitive specialization*: fine-tuning on a forward code transformation not only fails to produce the reverse but degrades the bidirectional reasoning a base model already had. Zhu et al. (2024) report the same shape in machine translation, where fine-tuning on a single direction generalizes to others *except* when English is on the target side, which causes task misinterpretation. We take the phenomenon as established in those settings. What is open is whether it is general, what predicts the size of the loss, how little reversed data prevents it, and whether what is lost is gone. Hasanov et al. (2026) evaluate 13 models on the duality of program-execution reasoning and find backward performance far below forward throughout, which is the premise this paper depends on: the capability is there to be destroyed.

The reversal curse: a capability never acquired. Berglund et al. (2024) fine-tune on "A is B" and find the reverse is not learned; Golovneva et al. (2024) cure it at the data level with reverse training, evaluated both data-matched and compute-matched. **The distinction is not rhetorical here, it is an experiment.** That literature describes a capability the model never had. Ours had it, and lost it. §7 measures relearning cost from a collapsed model against a *never-had control* — an invented transformation no pretraining corpus contains — and the two curves are what separate the cases. A second difference is budget: reverse training doubles the token count, while replacing forward pairs with their own reversal holds instances, sequence tokens and optimizer steps fixed at once.

Forgetting: diffuse loss, versus targeted and total. Fine-tuning degrades capabilities its data never threatened, across model scales (Luo et al., 2023), and LoRA forgets less than full fine-tuning (Biderman et al., 2024) — which makes the magnitudes we report on LoRA conservative, and is why §3 includes full fine-tuning arms. The difference from directional collapse is shape: forgetting is measured as a spread-out decline across unrelated benchmarks, where here the inverse direction goes to near zero while general probes move a few points. Scialom et al. (2022) show that a memory buffer of roughly 1 % of prior data preserves almost all earlier-task performance across eight sequential tasks. That is the same shape as our dose result and we do not claim the shape as new. Two things differ, and the second is testable: their buffer replays genuine earlier-task examples and *adds* to the training budget, where a reversed pair is constructed for free and *replaces*; and if directional collapse were ordinary forgetting, their result predicts that generic rehearsal at the same share should prevent it. REPLAY is that arm.

Suppression rather than removal, and skewed task inference. That fine-tuning modulates rather than replaces is already well supported outside this setting, and §7 does not claim it as new. Fine-tuning rarely alters underlying capabilities and learns a minimal "wrapper" over them (Jain et al., 2024); DPO does not remove the ability to produce toxic text but bypasses it with an offset distributed across layers (Lee et al., 2024) — a distribution that is itself a prediction for our layer ablation; and damaged capabilities can be recovered post hoc by filtering the weight update (spe, 2026). Closest of all is Kotha et al. (2024), who argue that models implicitly infer the task of a prompt and that fine-tuning *skews that inference* toward the fine-tuning distribution, recovering capability by making the prompt look farther from it. Our hypothesis is their mechanism with direction as the inferred task: forward-only training leaves the model applying the trained mapping whichever direction is asked for. What §7 adds is the directional case, a never-had control that makes erased-versus-suppressed a measurement rather than an inference, and four instruments that can disagree. lat (2026) add a caution we take seriously: objectives that repair reversal behaviour need not build a single direction-agnostic representation, and their probes are consistent with forward and reverse stored as distinct entries. That is why §7 uses four instruments rather than one, and why both directions share a single system prompt — if directions can be stored separately while behaviour looks bidirectional, two personas would let disjoint circuits masquerade as a unified one.

Direction as a confound in attribution. The methodological consequence has a literature of its own: gains attributed to an objective often belong to the data mixture (Longpre et al., 2023), and reporting practice makes such claims hard to check (Lipton and Steinhardt, 2018). Chen et al. (2025) is the sharpest current instance, and a useful one because nothing about it is careless. RevThink augments each example with a teacher-generated backward question and backward reasoning and improves *forward* accuracy by 13.53 % across 12 datasets. Its objective is $\mathcal{L} = \frac{1}{3n}\sum_i[\ell(\text{forward}) + \ell(\text{backward question}) + \ell(\text{backward reasoning})]$, a sum of cross-entropy terms over a union of three formats — which is what supervised fine-tuning on that union is. Its baselines differ from it in data direction as well as in method. When an objective has that form, the objective and the data mixture are not separable by any experiment the paper contains. That is the general problem being structural rather than accidental, and it is why the test in §1 is worth stating rather than assuming.

10 Conclusion

For any method that adds instances to paired training data, train the arm that supplies the same directional content and none of the method.

Limitations

Where the repair does not reach. ⟨⟨lim-invertibility⟩⟩

Scale. The full arm set runs at 1–4B with three seeds; only core arms run at 8–12B with two. ⟨⟨lim-scale⟩⟩

LoRA. Every arm is trained with LoRA, which maintains out-of-domain performance better than full fine-tuning (Biderman et al., 2024); the collapse magnitudes we report are therefore conservative, and the full fine-tuning arms in §3 bound the difference. ⟨⟨lim-lora⟩⟩

Resolution of the dose ladder. Adjacent rungs differ by a couple of points, which our evaluation sets cannot separate reliably. We therefore report the knee as a bound rather than a point estimate. ⟨⟨lim-knee⟩⟩

Criteria with a model in them. Two directions are scored by round-tripping through a frozen model rather than by execution. Each reports that model's own accuracy on the references as the criterion's ceiling. ⟨⟨lim-frozen-scorers⟩⟩

Languages and domains. The code cell is Python only; the translation cells cover two language pairs, both with English on one side. ⟨⟨lim-coverage⟩⟩

Acknowledgments

⟨⟨acknowledgments⟩⟩

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